

SKTLive × Kowri

Integration Options Technical Brief

22nd January 2026

Executive Summary

This document outlines two integration options for connecting SKTLive with Kowri's payment infrastructure. The recommended approach (Option 1) positions SKTLIVE as the system-of-record while leveraging Kowri's omni-channel payment capabilities. An alternative approach (Option 2) allows SKTLIVE to drive the integration by calling Kowri APIs directly.

Option 1: Kowri Integrates into SKTLive (Recommended)

In this model, SKTLIVE's platform remains the single source of truth for invoices, obligations, and payment status. Payer accesses any of Kowri's omni-channel collection points, Kowri performs the lookup and after a successful payment, Kowri sends payment notification via API to SKTLive.

Proven Track Record: We have successfully deployed this integration model for major clients including Meridian Port Services, Passion Air, NAGRAT, and Bank Hospital.

How It Works

The payer receives a payment reference from SKTLIVE (via SMS, email, or displayed in the SKTLIVE portal). They can then pay using any Kowri channel:

- **USSD Shortcode:** Dial ***227*xxx#** and enter the reference, or use embedded reference format
- **Web Payment Link:** Click a link sent via SMS or email from SKTLive (can also be converted to QR code)- <https://collections.kowri.app/xxx/xxxx>
- **Kowri App:** Available on iOS and Android
- **Kowri Business App:** For field agents to collect by entering/scanning the reference. The cash recording feature allows field agents to record cash collected for tracking purposes to provide a total representation of collections per agent.

- **Bank Channels:** Direct bank integrations (enabling in-branch and digital channel payments from Banks in the Kowri-Bank Network). We currently do this for MPS.
- **GHQR:** QR code payments (available end of Q1)

Embedded References

References can be embedded directly in the USSD shortcode or web link for auto-lookup. If embedded, the system performs an automatic lookup; otherwise, the user enters the reference manually on the channel.

USSD Example:

```
*227*130*2345678999#
```

*Format: *shortcode*MMDACode*LookupReference#*

Web Link Example:

```
https://collections.kowri.app/130/2345678999
```

This link can also be converted into a QR code for easy scanning.

Key Benefits

- **Centralized Ledger:** SKTLIVE maintains all business logic, invoices, taxpayer profiles, and revenue heads in one place.
- **Omni-Channel Collections:** Payer can pay through multiple channels without SKTLIVE managing each integration.
- **Reduced Complexity:** Payment edge cases (retries, timeouts, refunds) handled by Kowri.
- **Future-Proof:** New payment channels become automatic upgrades without SKTLIVE re-integration.
- **Field Collection Support:** Kowri Business App enables field agents to collect payments on-the-go.

Required SKTLIVE API Endpoints

SKTLIVE needs to provide endpoints that support the following three functions. The exact URL structure is flexible — we just need endpoints that deliver the described functionality:

Feature	Example Endpoint	Description
Authenticate	e.g. <code>/auth/token</code>	Secure authentication (OAuth2/JWT or API Key + HMAC)
Lookup Bill/Invoice	e.g. <code>/bills/lookup</code>	Return invoice/account details and amount due by reference
Payment Notification	e.g. <code>/payments/notify</code>	Receive payment confirmation with idempotency support

Authentication Options

- **Option A:** OAuth2 client credentials / JWT
- **Option B:** API Key + HMAC-SHA256 signing
- **Option C:** Mutual TLS (mTLS) for strongest transport-level control

Sequence Diagram

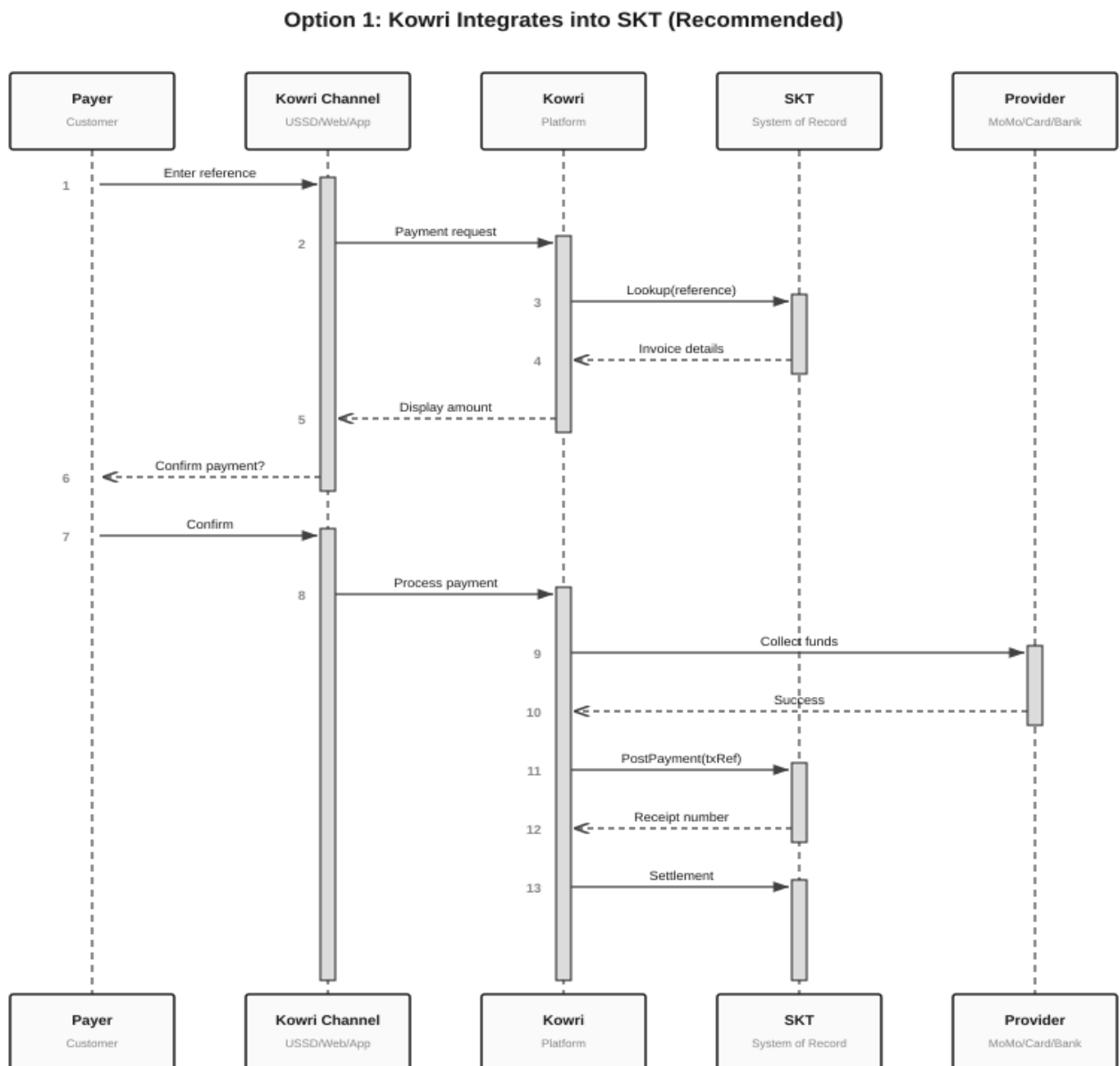


Figure 1: Payment Flow — Kowri Integrates into SKTLIVE (Recommended)

Error Handling

- **Idempotency:** SKTLIVE Payment Notification endpoint should accept an idempotency key (Kowri txRef) to prevent duplicate records.
- **Retries:** Kowri will retry payment notifications on transient errors; SKTLIVE should be tolerant of retries.
- **Reversals/Refunds:** MMDA can leverage on the bulk disbursement functionality on the merchant dashboard to process refunds and reversals. .

Option 2: SKTLIVE Integrates into Kowri APIs (Alternative)

In this model, SKTLive calls Kowri APIs directly to create invoices, initiate payments, check status, and receive callbacks when payments complete. This is the “plug-and-play / callback” model. This approach requires SKTLIVE to provide its payment channel for the MMDAs and will not use Kowri’s omni-channels.

Key Benefits

- **Direct Control:** SKTLIVE has more direct control over payment initiation logic within workflows.
- **Clear Separation:** SKTLIVE triggers payments, Kowri processes, SKTLIVE receives status via callback.

Required Kowri API Endpoints

For the simplest integration, SKTLIVE can use a single endpoint that combines invoice creation and payment initiation:

Endpoint	Method	Description
<code>/webpos/v2/payNow</code>	POST	Create invoice and initiate payment in one call
<code>/webpos/v2/checkPaymentStatus</code>	POST	Check current payment status
SKTLIVE Callback URL	POST	SKTLIVE-hosted endpoint to receive payment status updates

Security Model (Kowri V2)

All V2 requests require:

- **X-API-Key:** API key header for authentication
- **X-Signature:** HMAC-SHA256 signature using device secret
- **X-Request-Timestamp:** Timestamp within 5-minute window

Sequence Diagram

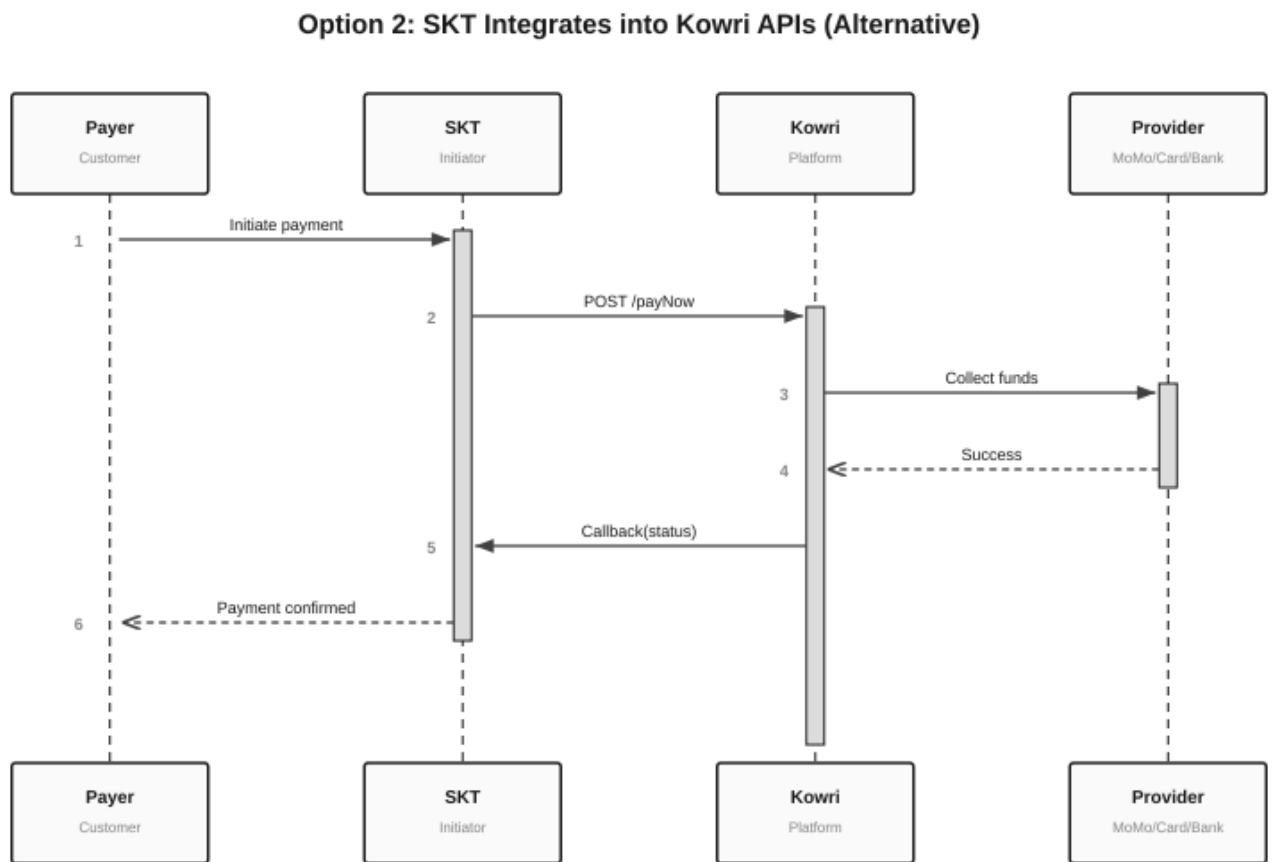


Figure 2: Payment Flow — SKTLIVE Integrates into Kowri APIs (Alternative)

Tradeoffs vs Recommended Approach

- SKTLIVE carries more integration surface area (invoice lifecycle, callbacks, status polling).
- When new channels are added, SKTLIVE may require additional testing per channel.

Our Recommendation

Proceed with Option 1 (Kowri integrates into SKTLIVE) first for the fastest, cleanest rollout across multiple MMDAs with SKTLIVE as the system-of-record. This approach leverages Kowri's full omni-channel capabilities including USSD, Web, App, Bank, and field agent collections — and has been proven with major clients like Meridian Port Services, Passion Air, NAGRAT, and Bank Hospital. Keep Option 2 available for specific MMDA edge cases or where SKTLIVE prefers to control invoice/payment initiation more tightly.

What We Need from SKTLIVE to Start

1. Technical Specification Document

- Authentication method for SKTLIVE APIs (JWT / API Key / mTLS)
- Lookup endpoint specification + sample response
- Payment notification endpoint specification + idempotency approach

2. Environment Access

- Sandbox/base URL
- Test credentials

3. Sample Test Data

- One MMDA sample dataset
- 5–10 invoice references / property IDs / BOP IDs for testing lookup and posting

Quick Comparison Summary

Aspect	Option 1 (Recommended)	Option 2 (Alternative)
System of Record	SKTLIVE	SKTLIVE
Integration Driver	Kowri calls SKTLIVE	SKTLIVE calls Kowri

SKTLIVE Development Effort	Minimal (3 endpoints)	Higher (callbacks, polling)
New Channel Support	Automatic	Requires testing
Edge Case Handling	Kowri manages	SKTLIVE manages
MMDA Scalability	High	Moderate

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